

Movies Recommendation System Project.

Improving Recommendation Accuracy through Model
Optimization

Problem Statement

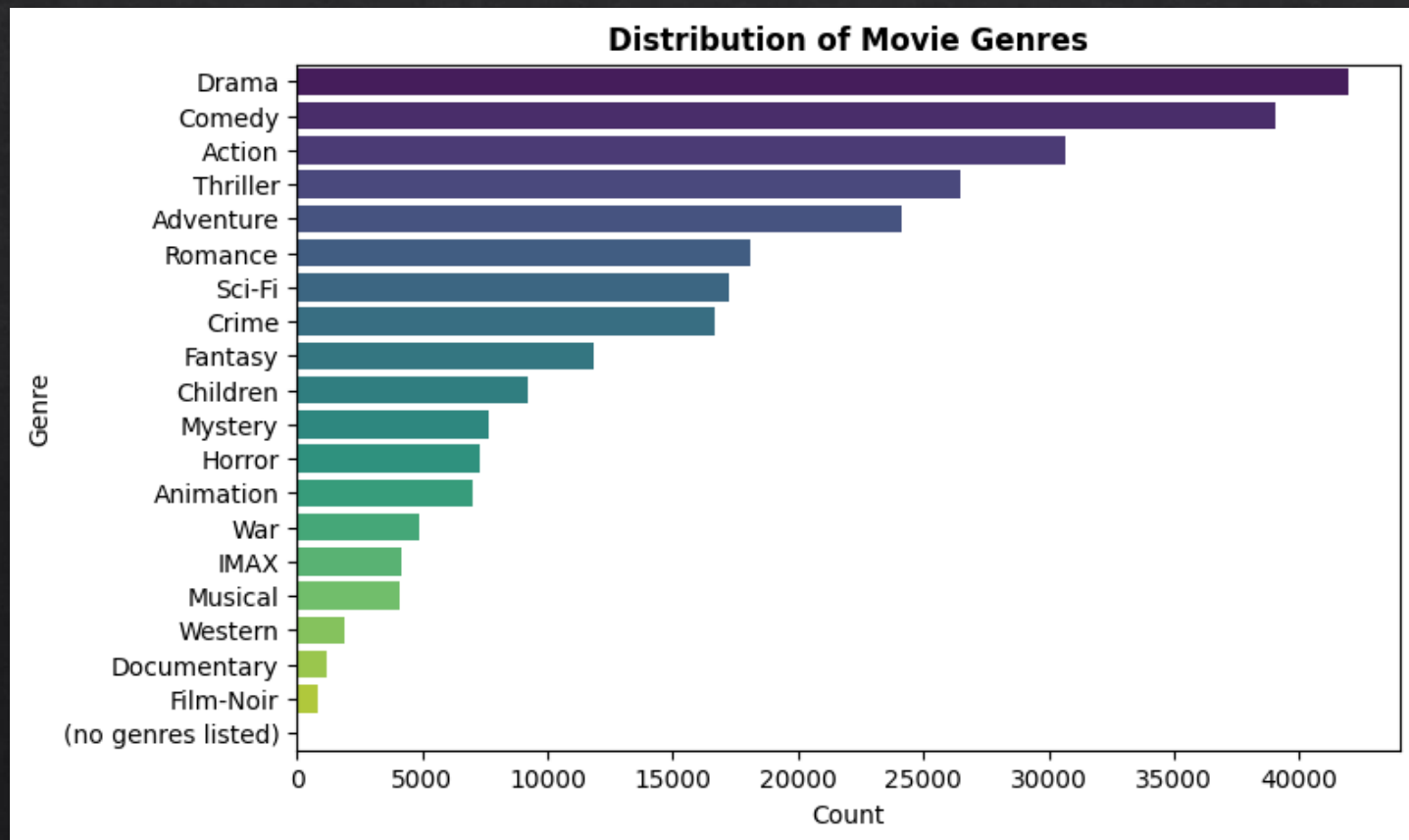
- ◆ Users struggle to discover relevant movies efficiently.
- ◆ Goal: Build a system that predicts user preferences accurately.

Objectives

- ◆ Better decision making using data driven insights.
- ◆ Build a movie recommendation system that suggests top 5 personalized movies to users.
- ◆ Analyze user ratings to understand preferences and similarities.

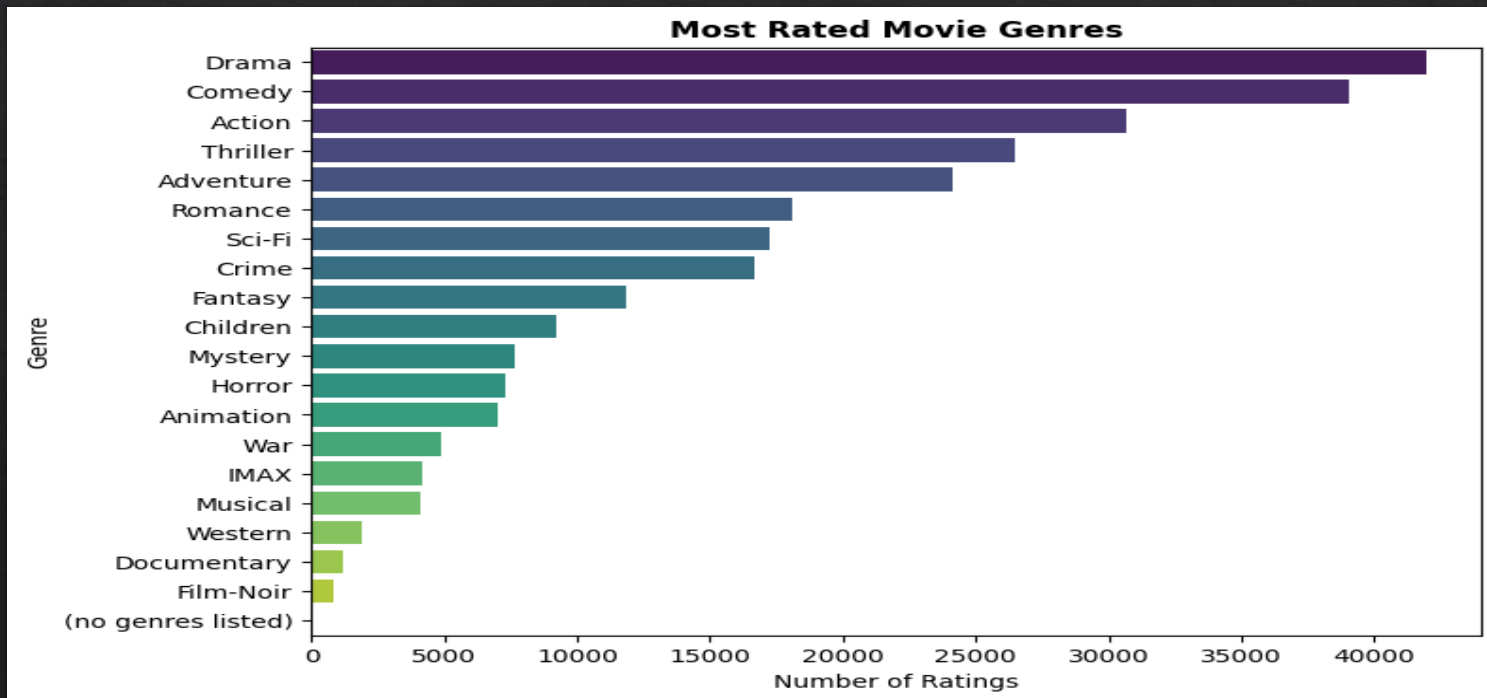
Distribution of movie genres

- Popular genres (Drama, Comedy, Action) dominate the dataset
- User interactions are concentrated in these categories
- May lead to biased recommendations toward popular genres



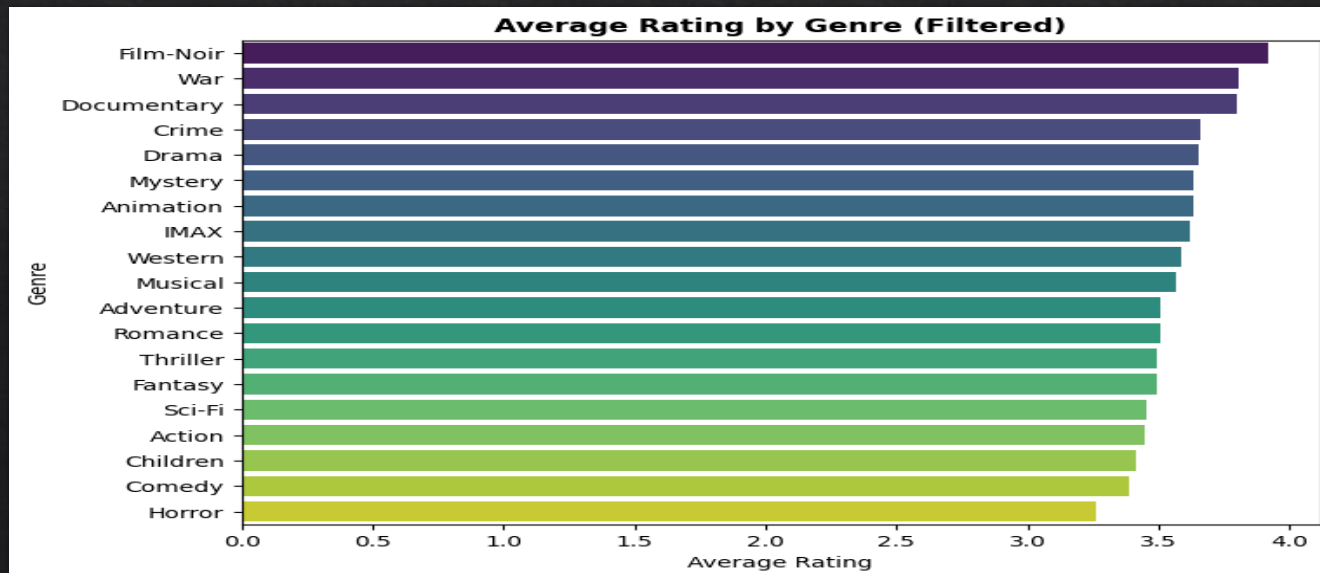
Most rated movie genre

Similar to the overall genre distribution, a few genres dominate user engagement, reinforcing the need to account for popularity bias when generating recommendations.



Average rating by genre

- **Highest-rated genres:** Film-Noir, War, and Documentary
- **Moderately rated genres:** Drama, Crime, Mystery, Animation
- **Lower-rated genres:** Comedy, Children, and Horror



Approach

- ◆ Used hybrid recommender function
- ◆ Used collaborative filtering (SVD model)
- ◆ Learned latent patterns between users and movies
- ◆ Compared baseline vs optimized model

Key Results

- ◆ Improved recommendation accuracy
- ◆ Better alignment with user preferences
- ◆ Demonstrated value of tuning process

Recommendations

- ◆ Implement hybrid recommendation system.
- ◆ Continuously collect user feedback.
- ◆ Monitor model performance.
- ◆ Address Data Sparsity-Integrate content based filtering techniques

Conclusions

- ◆ Developed a movie recommendation system using collaborative filtering, model tuning and hybrid techniques.
- ◆ Achieved strong RMSE performance and low MAE indicating reliable rating.
- ◆ Generated personalized Top-5 recommendations.
- ◆ Demonstrates real-world applicability and future potential.